

Lab 7: Find a procedure to transfer the files from one virtual machine to another virtual machine

Objective

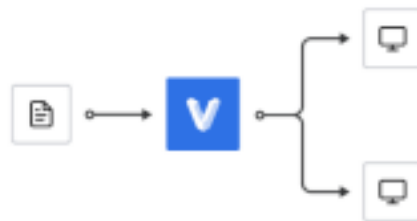
Find a procedure to transfer files from one virtual machine to another.

Tools Required

Tool	Purpose	Installation
VirtualBox	Hypervisor to create virtual machines	https://www.virtualbox.org/wiki/Downloads
Vagrant	Automates VM creation/configuration	https://developer.hashicorp.com/vagrant/downloads

What is Vagrant?

Vagrant is the command line utility for managing the lifecycle of virtual machines. Isolate dependencies and their configuration within a single disposable and consistent environment.



Part 1 — Creating a Local Virtual Network in VirtualBox

Step 1: Open VirtualBox

- Launch the **Oracle VM VirtualBox Manager**.
- Ensure no VMs are running initially.

Step 2: Create an Internal Network using Command Line

Open **Command Prompt** (Windows)

Command	Meaning
cd C:\Program Files\Oracle\VirtualBox\	Change Directory to C:\Program Files\Oracle\VirtualBox\

VBoxManage natnetwork add --netname testnet --network "192.168.56.0/24" --enable -- dhcp on	· --netname testnet → Names your network "testnet". · --network "192.168.56.0/24" → Creates a network range (subnet mask 255.255.255.0). · --enable → Activates the NAT network. · --dhcp on → Automatically assigns IPs to connected VMs.
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Screenshot for step 1 and step 2

```
C:\Users\harsh>cd C:\Program Files\Oracle\VirtualBox\

C:\Program Files\Oracle\VirtualBox>VBoxManage natnetwork add --netname testnet --network "192.168.56.0/24" --enable --dhcp on
VBoxManage.exe: error: NATNetwork server already exists
```

Step 3: Verify the Network

Command	Meaning
VBoxManage list natnetworks	Change Directory to C:\Program Files\Oracle\VirtualBox\

VBoxManage list natnetworks

You should see something like:

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\harsh>cd C:\Program Files\Oracle\VirtualBox\

C:\Program Files\Oracle\VirtualBox>VBoxManage natnetwork add --netname testnet --network "192.168.56.0/24" --enable --dhcp on
VBoxManage.exe: error: NATNetwork server already exists

C:\Program Files\Oracle\VirtualBox>VBoxManage list natnetworks
Name:          testnet
Network:       192.168.56.0/24
Gateway:       192.168.56.1
DHCP Server:   Yes
IPv6:          No
IPv6 Prefix:   fd17:625c:f037:2::/64
IPv6 Default:  No
```

Part 2 — Creating Virtual Machines and Attaching to the Network We'll use **Vagrant** to simplify VM setup.

Step 4: Create a Project Folder

Command	Meaning
cd C:\Users\arunv	Change Directory to create a project folder
mkdir IaaS_Lab1	create a project folder
cd IaaS_Lab1	Change directory



```

Vagrant.configure("2") do |config|
  # VM 1
  config.vm.define "vm1" do |vm1|
    vm1.vm.box = "ubuntu/bionic64"
    vm1.vm.hostname = "vm1"
    vm1.vm.network "public_network",
      bridge: "testnet",
      ip: "192.168.56.10"
    vm1.vm.provider "virtualbox" do |vb|
      vb.memory = "1024"
      vb.cpus = 1
      vb.name = "vm1"
    end
  end

  # VM 2
  config.vm.define "vm2" do |vm2|
    vm2.vm.box = "ubuntu/bionic64"
    vm2.vm.hostname = "vm2"
    vm2.vm.network "public_network",
      bridge: "testnet",
      ip: "192.168.56.11"
    vm2.vm.provider "virtualbox" do |vb|
      vb.memory = "1024"
      vb.cpus = 1
      vb.name = "vm2"
    end
  end
end

```

Figure 1 creating a Vagrant file

Explanation:

- Two VMs (vm1 and vm2) are created using Ubuntu images.
- Both are connected to the same **testnet** network.

Step 6: Launch the Virtual Machines

Run the following commands:

Command	Meaning
vagrant init	This will create a new environment and will create a new file name "Vagrantfile"

<pre>Vagrant.configure("2") do config # VM 1 config.vm.define "vm1" do vm1 vm1.vm.box = "ubuntu/bionic64" vm1.vm.hostname = "vm1" vm1.vm.network "public_network", bridge: "testnet",</pre>	Open the Vagrantfile in notepad. Delete all the contents. Copy and paste the code in the Vagrantfile as shown in figure 1
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<pre>ip: "192.168.56.10" vm1.vm.provider "virtualbox" do vb vb.memory = "1024" vb.cpus = 1 vb.name = "vm1" end end # VM 2 config.vm.define "vm2" do vm2 vm2.vm.box = "ubuntu/bionic64" vm2.vm.hostname = "vm2" vm2.vm.network "public_network", bridge: "testnet", ip: "192.168.56.11" vm2.vm.provider "virtualbox" do vb vb.memory = "1024" vb.cpus = 1 vb.name = "vm2" end end end</pre>	
<pre>vagrant up</pre>	This will: · Download the Ubuntu box (first time only) · Create and start both VMs · Attach them to the testnet network
<pre>vagrant ssh vm1 ip addr show vagrant ssh vm2 ip addr show</pre>	In command prompt type this command and notedown the ip address of both vm1 and vm2

```

PS C:\Users\harsh\IaaS_Lab1> vagrant ssh vm1
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-212-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue Sep 15 13:37:25 UTC 2026

System load:  0.14           Processes:            89
Usage of /:   2.9% of 38.70GB Users logged in:        0
Memory usage: 12%           IP address for enp0s3: 10.0.2.15
Swap usage:   0%            IP address for enp0s8: 192.168.56.10

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '20.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

vagrant@vm1:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 02:3b:7b:b7:3b:2d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86232sec preferred_lft 86232sec
    inet6 fd17:625c:f037:2:3b:7b:ff:feb7:3b2d/64 scope global dynamic mngtmpa
        valid_lft 86232sec preferred_lft 14232sec
    inet6 fe80::3b:7b:ff:feb7:3b2d/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d5:9b:82 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 brd 192.168.56.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 2401:4900:ccca:de4:a00:27ff:fed5:9b82/64 scope global dynamic mngtmpa
        valid_lft 7033sec preferred_lft 7033sec
    inet6 fe80::a00:27ff:fed5:9b82/64 scope link
        valid_lft forever preferred_lft forever

```

```

PS C:\Users\harsh\IaaS_Lab1> vagrant ssh vm2
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-212-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue Sep 15 13:38:38 UTC 2026

System load:  0.04          Processes:           90
Usage of /:   2.9% of 38.70GB Users logged in:        0
Memory usage: 12%          IP address for enp0s3: 10.0.2.15
Swap usage:   0%           IP address for enp0s8: 192.168.56.11

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '20.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

vagrant@vm1:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
fault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state U
P group default qlen 1000
    link/ether 02:3b:7b:b7:3b:2d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86232sec preferred_lft 86232sec
    inet6 fd17:625c:f037:2:3b:7bff:feb7:3b2d/64 scope global dynamic mngtma
ddr noprefixroute
        valid_lft 86232sec preferred_lft 14232sec
    inet6 fe80::3b:7bff:feb7:3b2d/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state U
P group default qlen 1000
    link/ether 08:00:27:d5:9b:82 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 brd 192.168.56.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 2401:4900:ccca:de4:a00:27ff:fed5:9b82/64 scope global dynamic mngt
mpaddr noprefixroute
        valid_lft 7033sec preferred_lft 7033sec
    inet6 fe80::a00:27ff:fed5:9b82/64 scope link
        valid_lft forever preferred_lft forever

```

Part 3 — Testing Network Connectivity

Step 8: Ping Between VMs

From inside **vm2**:

Command	You should see responses:
ping 192.168.56.10	<pre>vagrant@vm2:~\$ ping 192.168.56.10 PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data. 64 bytes from 192.168.56.10: icmp_seq=1 ttl=64 time=1.49 ms 64 bytes from 192.168.56.10: icmp_seq=2 ttl=64 time=0.444 ms 64 bytes from 192.168.56.10: icmp_seq=3 ttl=64 time=0.402 ms 64 bytes from 192.168.56.10: icmp_seq=4 ttl=64 time=1.25 ms 64 bytes from 192.168.56.10: icmp_seq=5 ttl=64 time=0.456 ms 64 bytes from 192.168.56.10: icmp_seq=6 ttl=64 time=0.395 ms 64 bytes from 192.168.56.10: icmp_seq=7 ttl=64 time=3.33 ms 64 bytes from 192.168.56.10: icmp_seq=8 ttl=64 time=1.06 ms 64 bytes from 192.168.56.10: icmp_seq=9 ttl=64 time=0.399 ms 64 bytes from 192.168.56.10: icmp_seq=10 ttl=64 time=0.395 ms 64 bytes from 192.168.56.10: icmp_seq=11 ttl=64 time=1.27 ms 64 bytes from 192.168.56.10: icmp_seq=12 ttl=64 time=1.45 ms 64 bytes from 192.168.56.10: icmp_seq=13 ttl=64 time=0.544 ms 64 bytes from 192.168.56.10: icmp_seq=14 ttl=64 time=0.729 ms 64 bytes from 192.168.56.10: icmp_seq=15 ttl=64 time=4.07 ms 64 bytes from 192.168.56.10: icmp_seq=16 ttl=64 time=6.17 ms 64 bytes from 192.168.56.10: icmp_seq=17 ttl=64 time=1.58 ms 64 bytes from 192.168.56.10: icmp_seq=18 ttl=64 time=2.70 ms 64 bytes from 192.168.56.10: icmp_seq=19 ttl=64 time=0.475 ms 64 bytes from 192.168.56.10: icmp_seq=20 ttl=64 time=1.49 ms 64 bytes from 192.168.56.10: icmp_seq=21 ttl=64 time=1.19 ms 64 bytes from 192.168.56.10: icmp_seq=22 ttl=64 time=7.25 ms 64 bytes from 192.168.56.10: icmp_seq=23 ttl=64 time=1.33 ms 64 bytes from 192.168.56.10: icmp_seq=24 ttl=64 time=0.515 ms</pre>

Expected Output:

```
vagrant@vm1:~$ scp sample.txt vagrant@192.168.56.11:/home/vagrant/
vagrant@192.168.56.11's password:
sample.txt                                100% 26   30.9KB/s   00:00
vagrant@vm2:~$ cat /home/vagrant/sample.txt
File transfer successful!
vagrant@vm2:~$ exit
logout
```

Both VMs can communicate over the internal network — meaning the **IaaS network setup** is successful.